

Restaurant Transaction Analysis and Expansion Recommendations for Foodie India

DOM207-DOM3007-DOM6401: MP1

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Data source: Foodie India proprietary restaurant transaction workbook

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Section: Introduction

Purpose

This project analyzes Foodie India's supplied restaurant transactions [1] to identify decision-relevant operating patterns and define reversible expansion pilots. It asks when higher-value dining occasions occur, how party size relates to per-head spend, what drives tip rate, and which operational tests are justified by the observed sample.

The Serve worksheet contains 365 records and seven supplied fields: Amount, Tip, Gender, Smoker, Day, Time, and Partysize. Findings are observational. The dataset contains no location, cost, margin, calendar date, capacity, footfall, or service-quality fields, so it cannot establish causation, profitability, site choice, or exposure-adjusted demand.

Executive findings

- **The pooled day effect is confounded with service period.** Lunch represents 78.4% of valid Thursday records but only 16.0% of valid Sunday records. Day distributions differ within lunch ($p=1.43e-05$) but not within dinner ($p=0.314$).
- **Weekend lunch is the expansion opportunity to test.** Saturday and Sunday lunch has $n=39$ and median bill 24.67, versus $n=89$ and 16.27 for Thursday and Friday lunch ($p=2.01e-06$).
- **Per-head spend declines as party size rises.** Spearman $\rho=-0.197$ ($p=2.24e-04$); median per-head spend is 8.22 for two-person parties and 6.43 for four-person parties.
- **Tip rate declines as bill amount rises.** After removing 3 probable tip-entry errors, $\rho=-0.433$ ($p=7.12e-18$). Tip rate does not differ by waiter gender, smoker status, service period, or day after false-discovery-rate control.

Decision posture: all recommendations are controlled pilots with stop rules. None is a permanent capacity, compensation, or site decision.

Section: Data (pre)processing

Cleaning and traceability

1. The script validates the seven required columns, copies the source data, and adds SourceRow. The workbook is never edited.
2. Obvious decimal-format defects in Amount and Tip are repaired; non-positive bills and missing or negative tips are excluded only from monetary analysis.
3. Unambiguous category variants are normalized. San and Sn map to Sunday; ambiguous S, SS, and SSS remain missing.
4. Integer party sizes 1-6 define per-head analysis. Per-head spend equals bill divided by party size.
5. Excel rows 303-310 are flagged as DuplicateBlock but retained. SourceRow 296 is separately flagged TipSuspect and retained. Tips exceeding three times a positive bill are flagged as TipEntryError.

Data-quality audit

Check	Records	Treatment
Source records	365	Retained; source workbook unchanged
Exact duplicate-looking rows (7 exact; 8-row pasted block at Excel rows 303-310)	7	All retained; pasted copy flagged as DuplicateBlock
Amount text formats repaired	3	Comma/internal hyphen converted to decimal
Tip text formats repaired	1	Trailing backtick removed
Categorical values normalized	65	Only unambiguous variants mapped, including San/Sn to Sun
Ambiguous/missing day	11	Excluded only from day analysis
Ambiguous/missing time	4	Excluded only from time analysis
Ambiguous/missing smoker status	9	Excluded only from smoker analysis
Non-positive/unparseable amount	1	Excluded from monetary analysis
Missing/negative tip	2	Excluded from tip analysis
Analysis-ready amount/tip records	362	Primary analytical base
Probable tip entry errors	3	Tip exceeds three times bill; excluded from tip-rate versus bill test
Suspect original-block tip	1	SourceRow 296 has tip 13.55; retained in primary analysis and tested in sensitivity
Other flagged tip rates above 100%	4	Retained; flagged but not asserted as errors
Core party-size records (1-6)	345	Used for party-size and per-head analysis
Pasted-block sensitivity sample	354	Drops SourceRow 303-310 only; primary conclusions re-estimated

Counts are not additive because one row may meet more than one condition.

Section: Data (pre)processing - anomaly review

Pasted-block finding

Excel rows 295-302 reappear as rows 303-310. The copy is the cleaner version: original row 296 contains the suspect tip 13.55 (about 38% of a 35.60 bill), whereas copy row 304 contains 3.55. The seven exact duplicates comprise 5 exact rows inside the pasted block plus 2 unrelated exact pairs at Excel rows 198/199 and 254/316; 3 additional block rows differ slightly. Without a transaction ID, all records remain in the primary analysis.

Dropping the eight copied rows gives n=354. The fourth sensitivity also drops TipSuspect. Pooled day and correlation conclusions remain unchanged across all four scenarios.

Scenario	n	Pooled day p	Bill-tip rho	Bill-tip-rate rho
Baseline	362	0.003	0.639	-0.422
Drop DuplicateBlock	354	0.003	0.643	-0.413
Drop TipEntryError	359	0.003	0.643	-0.433
Drop DuplicateBlock and TipSuspect	353	0.003	0.640	-0.423

Probable tip-entry errors

Three recorded tips exceed three times their positive bill and are probable decimal-loss errors. They remain in the audit file but are excluded from the primary bill-versus-tip-rate test and Figure 4.

Excel row	Bill	Recorded tip	Recorded rate
91	21.16	300.00	1,417.8%
105	22.42	3487.00	15,553.1%
232	22.12	288.00	1,302.0%

Another 4 rates lie between 100% and 300%. They remain flagged but are not asserted to be errors. Restricting the association to rates at or below 100% gives n=355, rho=-0.429, and p=2.69e-17, so the negative relationship is robust.

Section: Data visualization

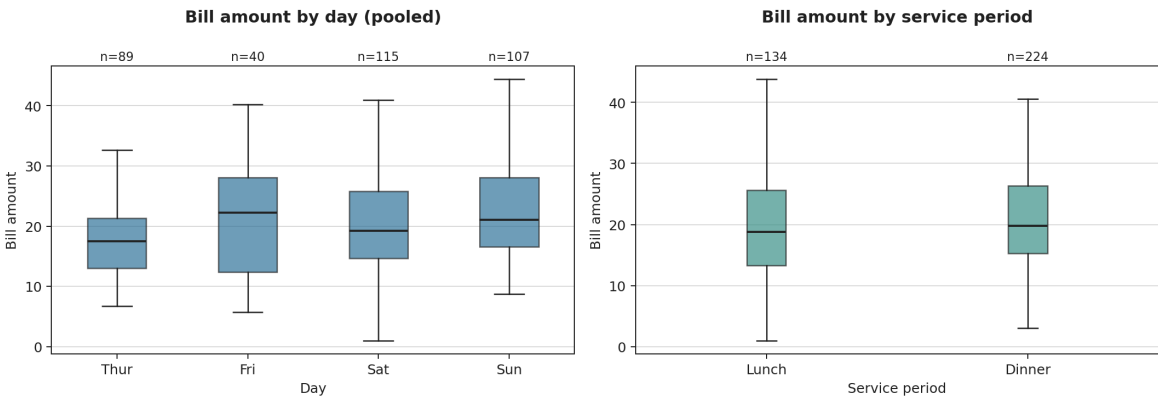


Figure 1. Boxplots compare pooled bill distributions by day and service period; n labels sit above the plotting areas. The pooled day panel uses n=351; the stratified day-by-service analysis uses n=347 because 4 valid-Day rows lack valid Time. For legibility, 14 day-panel and 11 service-panel plotting outliers are hidden under the 1.5 x IQR rule; all observations remain in tests.

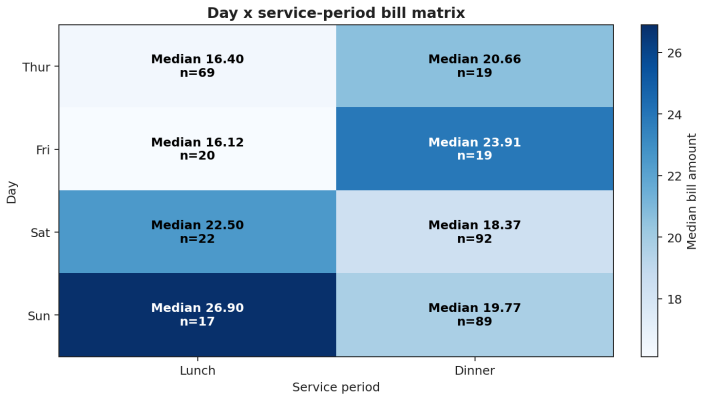


Figure 2. Heatmap of median bill and n for the n=347 records with valid Day and Time. Weekend lunch versus weekday lunch is the decision contrast. Lunch cells are small: Friday n=20, Saturday n=22, and Sunday n=17.

Data visualization: decision measures

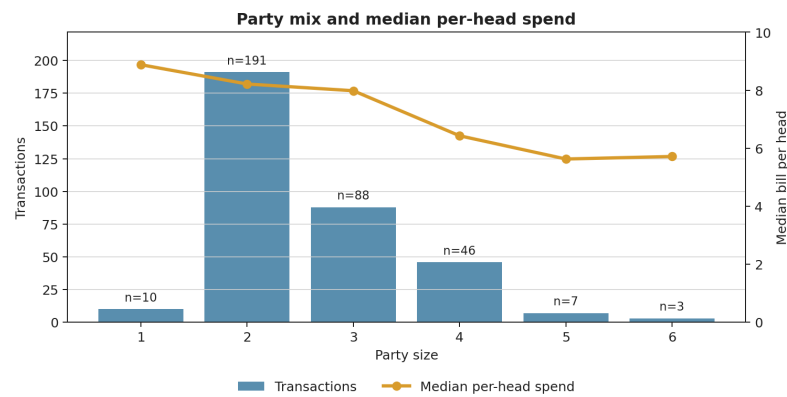


Figure 3. Bars show transaction n and the line shows median per-head spend for party sizes 1-6, with the per-head axis starting at zero. Party sizes 5 and 6 have only n=7 and n=3.

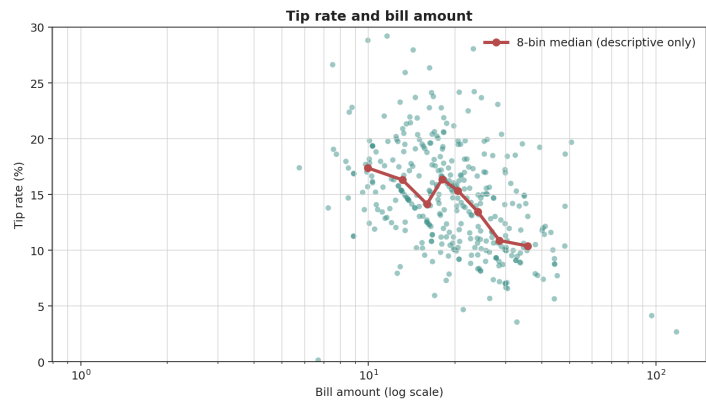


Figure 4. Scatterplot of tip rate versus bill on a logarithmic bill axis with an 8-bin median trend. The 3 probable entry errors are removed. The y-axis is capped at 30%; 7 valid points above the cap have rounded rates 33, 38, 65, 104, 112, 118, 166%. The trend is descriptive only.

Section: Analysis

Day findings must be interpreted within service period

The pooled day test and Figure 1 use every analysis-ready row with valid Day (n=351). The within-period tests and day-by-service table require valid Day and Time (n=347); the two bases differ because 4 valid-Day rows have missing or ambiguous Time.

The pooled day test is significant (Kruskal-Wallis H=13.841, p=0.003, epsilon-squared=0.031). However, lunch exposure differs sharply by day:

Day	Valid n	Lunch n	Dinner n	Lunch share
Thur	88	69	19	78.4%
Fri	39	20	19	51.3%
Sat	114	22	92	19.3%
Sun	106	17	89	16.0%

After stratification, lunch days differ (H=25.158, p=1.43e-05, epsilon-squared=0.179), while dinner days do not (H=3.554, p=0.314, epsilon-squared=0.003). Weekend lunch has median bill 24.67 (n=39) versus 16.27 (n=89) for weekday lunch; rank-biserial effect=0.529, p=2.01e-06. Thursday lunch and Friday lunch do not differ (p=0.709). Thus the former pooled Thursday-Sunday interpretation is primarily a lunch/dinner composition effect.

For core-party lunch rows, weekend per-head spend has median 9.85 (n=36) versus 7.26 (n=83) on weekdays; C7 gives p=7.32e-05 and rank-biserial effect=0.459. Mean party size is 2.69 versus 2.36, so the higher weekend per-head result is not explained by smaller parties.

Methods and multiplicity

Rank-based procedures are used because monetary variables are skewed: Mann-Whitney U [2], Kruskal-Wallis [3], and Spearman rho [4]. Tests are two-sided. Benjamini-Hochberg adjustment [6] is applied separately to the eight primary and seven context tests; lunch post-hoc comparisons are adjusted separately. Effect sizes accompany p-values under ASA guidance [5], with the H-statistic epsilon-squared convention [9]. Calculations use SciPy [8].

ID	Primary test	n	Statistic	Raw p	BH p	Effect	Decision
H1	Bill amount by day within lunch	128	25.158	1.43e-05	3.82e-05	0.179	Reject H0
H2	Bill amount by day within dinner	219	3.554	0.314	0.418	0.003	Do not reject H0
H3	Weekend vs weekday lunch bill	128	2,654.000	2.01e-06	8.04e-06	0.529	Reject H0
H4	Party size vs per-head spend	345	-0.197	2.24e-04	4.48e-04	-0.197	Reject H0
H5	Bill amount vs tip rate	359	-0.433	7.12e-18	5.70e-17	-0.433	Reject H0
H6	Tip rate by waiter gender	362	16,263.000	0.802	0.802	0.015	Do not reject H0
H7	Tip rate by smoker status	353	16,249.500	0.190	0.304	0.082	Do not reject H0
H8	Tip rate by service period	358	14,434.000	0.545	0.623	-0.038	Do not reject H0

Analysis: commercial interpretation

Per-head spend

Party size	n	Median bill	Median per head
1	10	8.88	8.88
2	191	16.43	8.22
3	88	23.95	7.98
4	46	25.73	6.43
5	7	28.15	5.63
6	3	34.30	5.72

Per-head spend declines with party size ($\rho=-0.197$, $p=2.24e-04$). Two-person median is 8.22, compared with 6.43 for four-person parties; the exploratory two-versus-four comparison gives $p=4.85e-04$ and $\text{effect}=0.332$. The positive party-size-versus-total-bill correlation is context only and does not show that larger groups are more productive per seat.

Tip rate

Bill quartile	n	Minimum bill	Maximum bill	Median tip rate
Q1 (lowest)	90	1.01	14.31	16.9%
Q2	90	14.48	19.10	15.8%
Q3	89	19.12	25.71	14.3%
Q4 (highest)	90	25.89	117.81	10.6%

Median tip rate falls from 16.9% in the lowest bill quartile to 10.6% in the highest. The primary association is $\rho=-0.433$, $p=7.12e-18$. Tip rate does not differ by waiter gender ($p=0.802$), smoker status ($p=0.190$), service period ($p=0.545$), or day ($p=0.331$).

The decline is partly a **flat-dollar** tip effect: median tip is 2.87, and 144 of 359 tips (40.1%) are multiples of 0.50. Tip rate is negatively associated with per-head spend ($\rho=-0.455$, $p=7.61e-19$) but not clearly with party size ($\rho=-0.095$, $p=0.080$). Therefore the bill-rate pattern does not show that larger spenders tip less generously. Bill and tip amount $\rho=0.639$ remains mechanical context.

Section: Discussions and Recommendations

R1. Weekend lunch service pilot

Justification: Saturday and Sunday lunch has median bill 24.67 ($n=39$). Core-party weekend lunch also has higher median per-head spend (9.85, $n=36$) than weekday lunch (7.26, $n=83$), with $p=7.32e-05$; mean party size is larger, not smaller (2.69 versus 2.36).

Action: test an extended or added weekend lunch window on matched dates with comparable untreated shifts.

Success metric: incremental covers, labour hours, gross profit per available seat-hour, median bill, and complaints versus control.

Stop rule: stop if gross profit per available seat-hour does not improve after added labour, or complaints materially worsen.

R2. Thursday lunch value-per-head pilot

Justification: Thursday lunch has the largest lunch cell ($n=69$) and lowest observed lunch per-head median (6.85, $n=63$ valid core-party rows).

Action: test one transparent add-on or meal-upgrade prompt on matched Thursday lunch shifts; do not scale on volume alone.

Success metric: per-head spend, attachment rate, gross profit per seat-hour, and complaints.

Stop rule: stop if per-head spend and gross profit per seat-hour do not improve together or complaints increase.

R3. Small-party and flexible-table pilot

Justification: median per-head spend is 8.22 for two-person and 6.43 for four-person parties; the overall association is negative. Large-party bundles are not supported.

Action: favour two-top tables that pair flexibly. Treat any three-to-four-person add-on as a measured experiment.

Success metric: per-head spend, seat utilization, turn time, and gross profit per seat-hour.

Stop rule: revert if utilization or gross profit per seat-hour falls without a guest-service benefit.

Limitations

- Sampling frame, collection dates, transaction independence, and current representativeness are unknown.
- Lunch cells are small: Friday $n=20$, Saturday $n=22$, Sunday $n=17$.
- Party size may include children or shared dishes; per-head spend is an approximation of per-diner spend.
- No cost, margin, discount, labour, capacity, location, or footfall fields exist.
- No transaction ID exists; the pasted block cannot be conclusively classified.
- All results are observational and do not establish causation.

Interpretation: service-period composition prevents a standalone Sunday or Thursday action. Pilots must compare like-for-like shifts.

Recommendations: controls and boundaries

R4. Tip-policy control

Justification: tip rate declines as bills rise; no adjusted difference is established by waiter gender, smoker status, service period, or day. Hospitality evidence warns that raw tip size is a weak service-quality measure [7].

Action: do not rank, schedule, or pay staff using raw tips. Evaluate any service coaching with guest feedback and rate-based measures.

Success metric: guest ratings, complaints, tip-rate distribution, and staff dispersion after controlling for bill mix.

Stop rule: withdraw incentives that create demographic segmentation, unstable rankings, or no guest-outcome improvement.

R5. Data-capture prerequisite

Justification: the workbook cannot distinguish sites, dates, repeat records, costs, margins, capacity, or exposure.

Action: add timestamp, transaction ID, location, menu/category, discount, direct-cost, labour-hour, table-capacity, and available-seat-hour fields.

Success metric: complete identifiers and exposure/cost fields for every pilot transaction, with reconciliation checks.

Stop rule: do not approve permanent expansion while transaction identity, cost, and capacity exposure remain unavailable.

Actions not supported by the data

- No city, market, or site recommendation.
- No permanent capacity addition based on transaction counts.
- No four-person bundle as a default proposition.
- No staffing or compensation decision based on raw tips, waiter gender, or smoker status.
- No causal claim about spend, tips, satisfaction, or retention.
- No profit or return-on-investment forecast.

Section: References

[1] Foodie India. (n.d.). *Restaurant.xlsx* [Proprietary transaction dataset] and *DOM207-DOM3007-DOM6401: MP1* [Project specification]. Supplied course files; accessed 21 September 2026.

[2] Mann, H. B., and Whitney, D. R. (1947). "On a Test of Whether One of Two Random Variables Is Stochastically Larger than the Other." *Annals of Mathematical Statistics*, 18(1), 50-60. <https://doi.org/10.1214/aoms/1177730491>.

[3] Kruskal, W. H., and Wallis, W. A. (1952). "Use of Ranks in One-Criterion Variance Analysis." *Journal of the American Statistical Association*, 47(260), 583-621. <https://doi.org/10.1080/01621459.1952.10483441>.

[4] Spearman, C. (1904). "The Proof and Measurement of Association between Two Things." *The American Journal of Psychology*, 15(1), 72-101. <https://doi.org/10.2307/1412159>.

[5] Wasserstein, R. L., and Lazar, N. A. (2016). "The ASA's Statement on p-Values: Context, Process, and Purpose." *The American Statistician*, 70(2), 129-133. <https://doi.org/10.1080/00031305.2016.1154108>.

[6] Benjamini, Y., and Hochberg, Y. (1995). "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society: Series B*, 57(1), 289-300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.

[7] Lynn, M., and McCall, M. (2000). "Gratitude and Gratitude: A Meta-Analysis of Research on the Service-Tipping Relationship." *Journal of Socio-Economics*, 29(2), 203-214. [https://doi.org/10.1016/S1053-5357\(00\)00062-7](https://doi.org/10.1016/S1053-5357(00)00062-7).

[8] SciPy Community. (2026). *scipy.stats API Reference: mannwhitneyu, kruskal, spearmanr, and false_discovery_control*. <https://docs.scipy.org/doc/scipy/reference/stats.html>; accessed 21 September 2026.

[9] Kassambara, A. (2023). *rstatix: Kruskal-Wallis Effect Size*. R package documentation. Formula $\eta^2[H] = (H - k) / (n - k)$. https://search.r-project.org/CRAN/refmans/rstatix/html/kruskal_effsize.html; accessed 21 September 2026.

Reproducibility statement

Run `python analysis.py` to regenerate cleaned data, audit tables, tests, sensitivities, and figures. Then run `python verify.py` to check source-row preservation, flags, test families, figures, report content, and landscape layout.